Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the HDI Predictor can be scaled to handle larger user bases, increased data loads, and extended features. It also captures the roadmap for enhancing the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority
1	Training dataset is small (53 countries). Values outside the training range may give inaccurate predictions.	Reduced model accuracy for uncommon or extreme input values.	High
2	Model retrains every time app.py restarts. No saved model file (.pkl) used.	Startup delay of a few seconds before the server is ready.	Medium
3	No server-side input validation. Out-of-range values are accepted without rejection.	Unreliable predictions if users enter invalid numeric values.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded Flask dev server. Cannot handle multiple users simultaneously.	Deploy with Gunicorn (multiple workers) on Render or AWS. Add a load balancer for high traffic.
2	Data Storage	Flat CSV file with 53 rows. Loaded entirely into memory at startup.	Migrate to PostgreSQL or SQLite with SQLAlchemy ORM. Supports thousands of countries.
3	Performance	Model trained fresh on every restart. No caching or persistence layer.	Save trained model using joblib.dump(). Load at startup with joblib.load() to skip retraining.
4	Security	No authentication, no HTTPS enforcement, no numeric range validation.	Add Flask-Login for sessions. Enforce HTTPS on Render. Add server-side input range checks.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Auto-fill feature — selecting a country from the dropdown auto-populates all four input fields with real values from the dataset.	1–2 months	Improves UX and reduces manual data entry errors.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	Expand dataset to 150+ countries using live UNDP HDI data API. Retrain and persist model with joblib.	2–3 months	Higher accuracy and broader global coverage across all tiers.
Phase 4	Add data visualizations — bar charts, scatter plots, and world map choropleth showing HDI scores using Plotly.	3–4 months	Transforms the app into a full HDI analytics dashboard.